Design and Development of a Novel Electro-Mechanical Mouth System for Food Oral Processing Research

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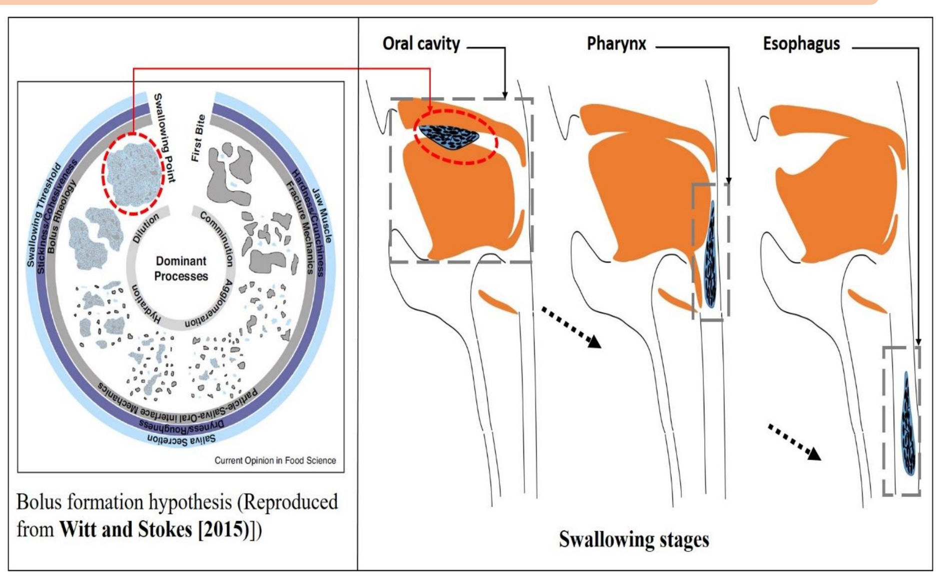
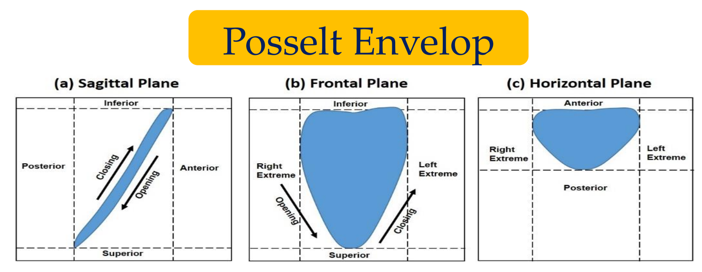
INTRODUCTION

Chewing Process Hypothesis/Models:

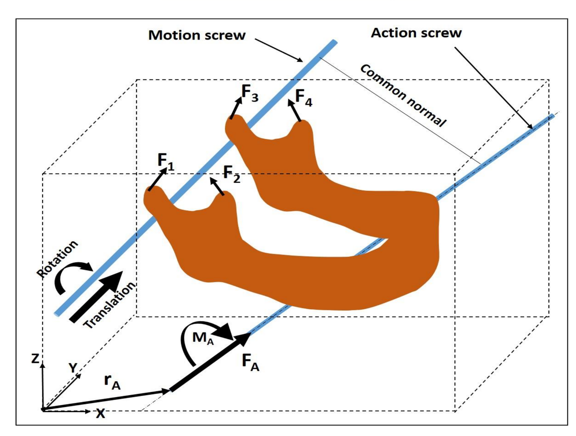
Eating Process Chain from Intake to Swallowing



Posselt Envelop



Screw Theory



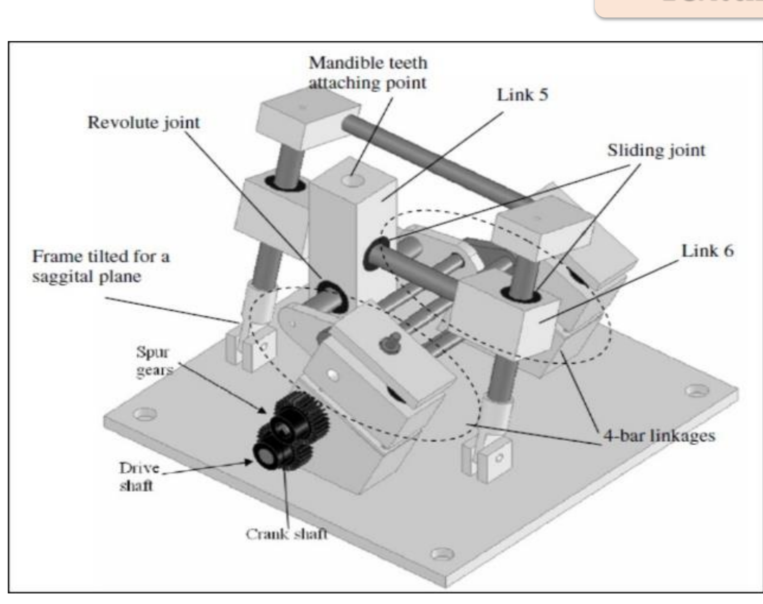
Eating Process Concomitance/Chain

Intake → Breaking-down → Saliva mixing → Building-up ("Bolus Formation") → *Swallowing*

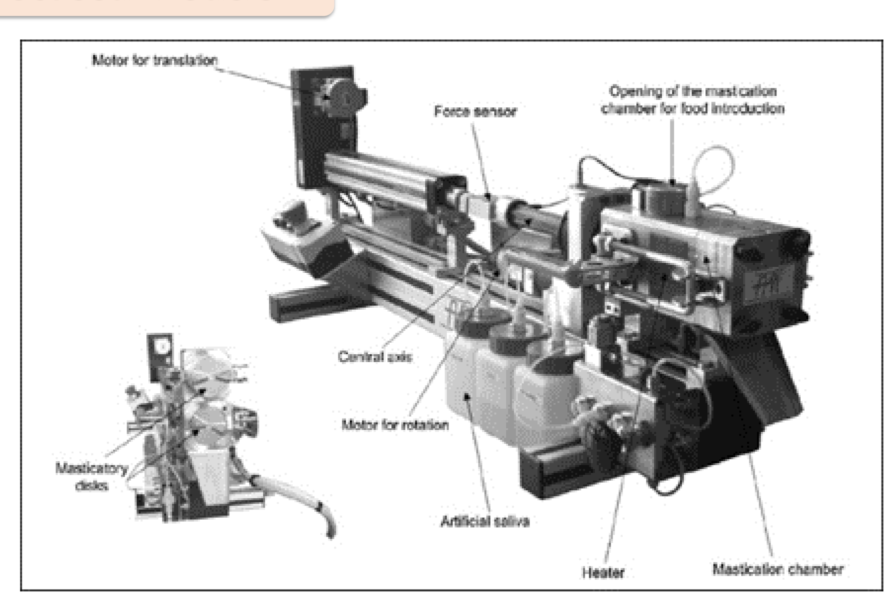
Instrumental Simulation

- Food textures and flavor studies are developed as distinct disciplines in food oral processing research
- Instruments developed so far have noted limitations which are also well justified
- Attempted studies either focused on chewing simulation (or texture assessment) or for flavor detection

Texture Focused Models

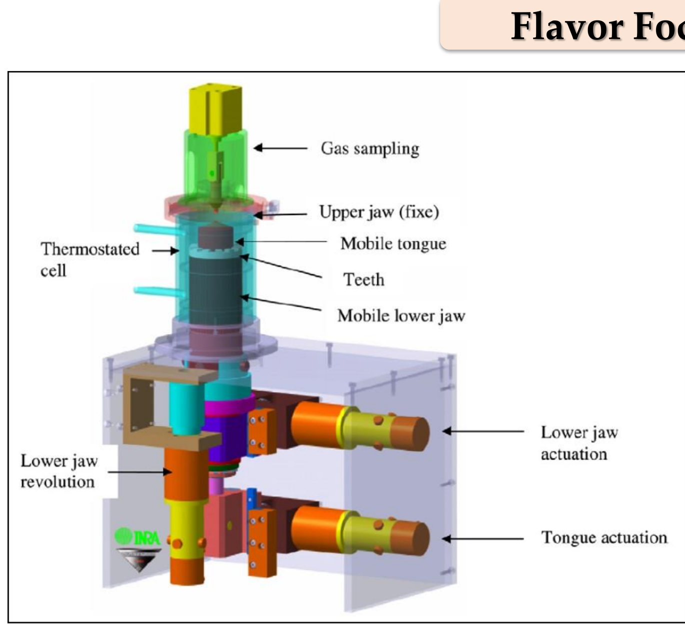


(Xu, Lewis, et al., 2008)

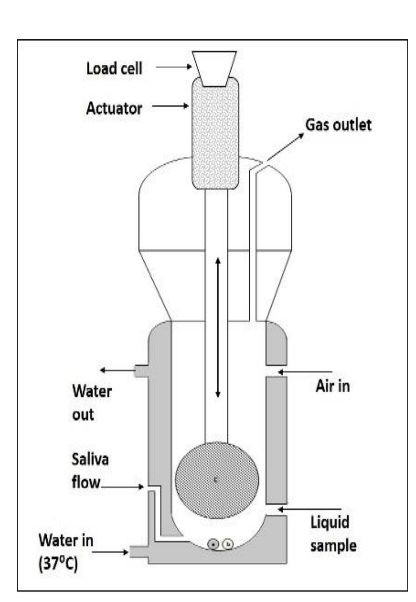


(Woda *et al.*, 2010; Mishellany-Dutour *et al.*, 2011)

Flavor Focused Models



(Salles *et al.*, 2007)



(Benjamin et al., 2012)

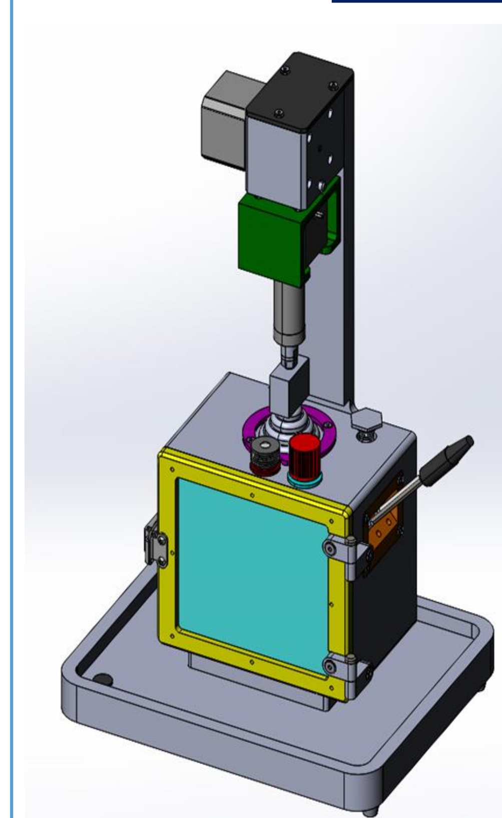
CHALLENGES

- Inter-individual variations: **chewing pattern**; **saliva secretion and composition; psycho-physical factors.**
- In-process manipulation of food by tongue (it's hard to mimic the **exclusive mechanical flexibility, surface structure,** and it's **neuro-sensory control system**!).
- Materials for the design (must be **inert** and **low absorbent of gaseous compounds**).

OBJECTIVES

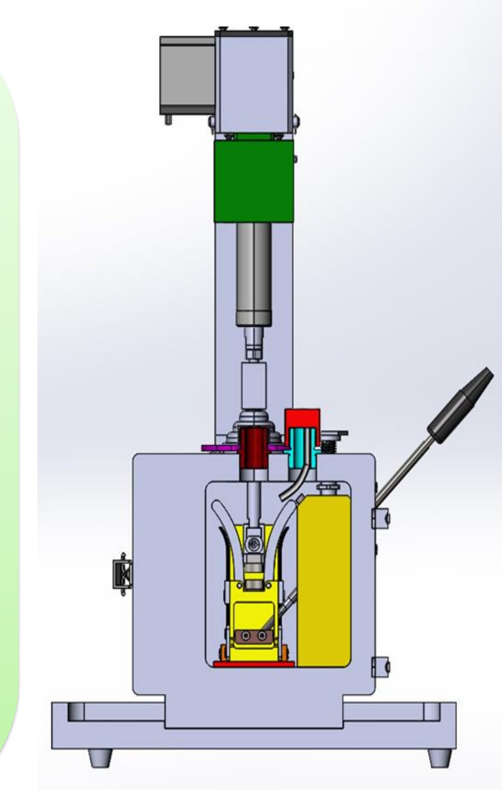
- To design and develop a *comprehensive* instrumental system for food oral processing research
- To assess *"food texture"* and *"food flavor"* characteristics simultaneously

SALIENT FEATURES OF THE DESIGN



- Bi-directional jaw movement
- Hermetically sealed chamber with temperature and volume control features
- Realization of actual geometry of the molar teeth

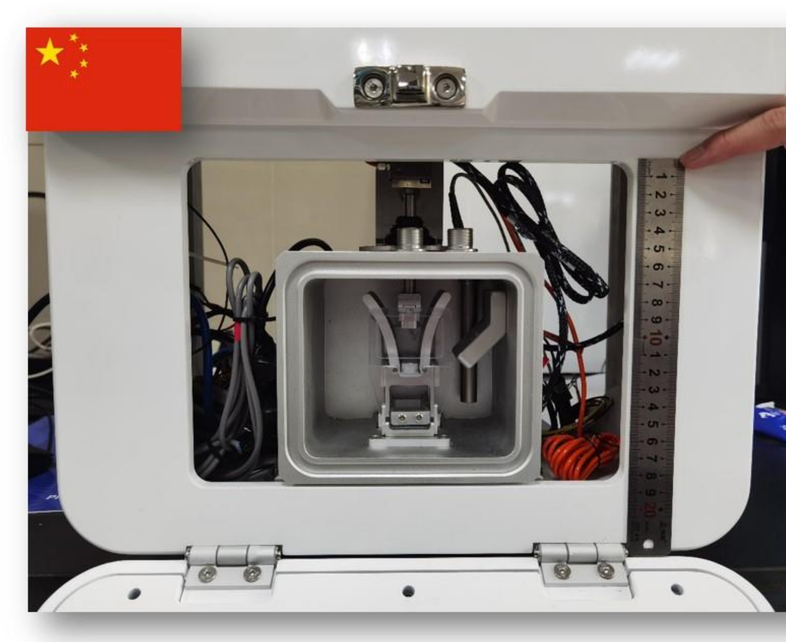
- Twin saliva injection points
- Facilities to collect flavor release compounds
- In-situ recording of chewing force, trajectory, pH, conductivity



- Extraction facility for food/saliva mixture for ex-situ analyses
- Easily replaceable/interchangeable sub-assembly and components
- User friendly GUI control, easy cleaning and lean maintenance

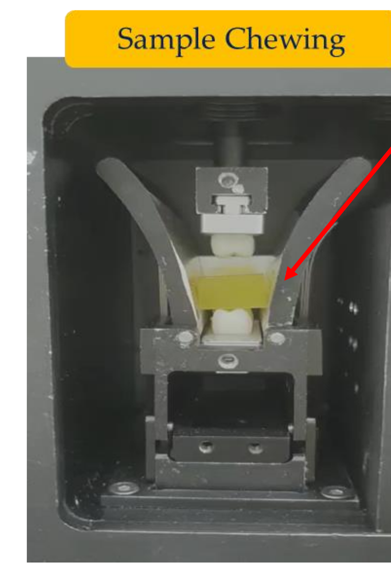
THE FIRST PROTOTYPES

Twins just delivered in Israel and China!

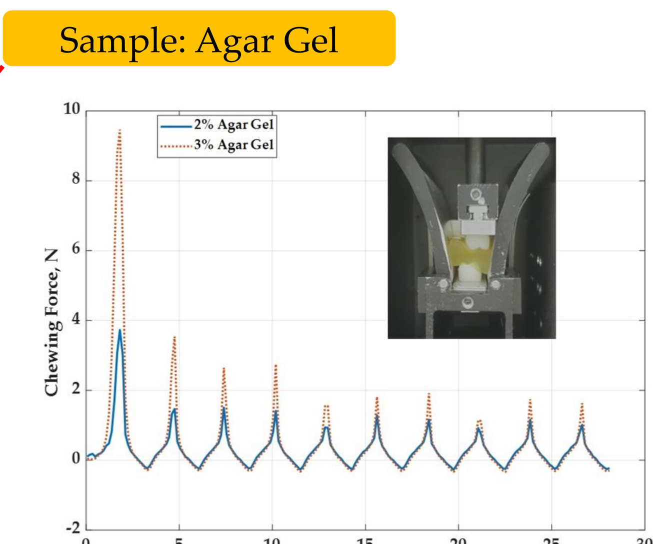


PRELIMINARY RESULTS

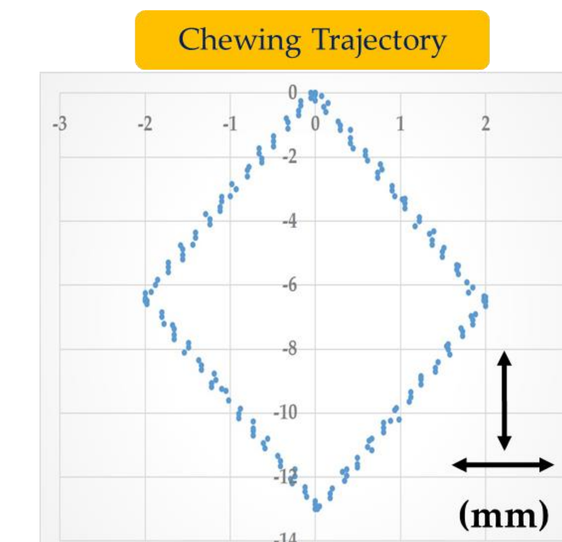
Sample Chewing



Sample: Agar Gel



Chewing Trajectory



CONCLUSIONS

- Based on the present state of development and preliminary experimental trials, we found these novel instrumental progresses as highly promising to bring in comprehensive characterization tools and methods for food product assessment.
- In addition, we found exciting scopes, to be addressed in more details in nearer time, which will offer greater instrumental advancement for both academics and industry on the topics of food oral processing research.

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1. Chen, J. (2009) 'Food oral processing—A review', *Food Hydrocolloids*. Elsevier, 23(1), pp. 1–25
2. Witt, T. and Stokes, J. R. (2015) 'Physics of food structure breakdown and bolus formation during oral processing of hard and soft solids', *Current Opinion in Food Science*
3. Sandip Panda, Jianshe Chen, Ofir Benjamin, *Development of Model Mouth for Food Oral Processing Studies: Present Challenges and Scopes,* Innovative Food Science and Emerging Technologies, 66, December, **2020**

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